

# CLOVRR STEM LEARNING

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## K-8 DIAGNOSTIC EXAM: AMC 8 & PRE-ALGEBRA READINESS

Time Limit: 30 Minutes | Target Grade: 4-8 | No Calculators Allowed

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### Section 1: Arithmetic Fluency & Number Theory

1. What is the value of  $(25 \times 16) - (25 \times 12)$ ? *(Hint: Use the distributive property instead of calculating the large numbers!)*
  2. The sum of three consecutive odd integers is 105. What is the largest of these integers?
  3. What is the greatest common factor (GCF) of 48 and 72?
  4. If  $x = 3$  and  $y = -2$ , evaluate  $x^2 - 3xy + y^2$ .
  5. A recipe requires  $\frac{3}{4}$  cup of sugar for every 2 cups of flour. If you use 6 cups of flour, how much sugar do you need?
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### Section 2: Algebra & Logical Mechanics

6. Solve for  $x$ :  $3(x - 4) + 5 = 2x - 1$
  7. A train leaves Dallas traveling at 60 mph. Another train leaves Houston (240 miles away) at the same time, traveling toward Dallas at 40 mph. How many hours until they meet?
  8. The length of a rectangle is 4 inches more than twice its width. If the perimeter is 44 inches, what is the width?
  9. Simplify the following expression:  $\frac{4x^3 y^2}{2x y^4}$
  10. If  $5^{x+2} = 125$ , what is the value of  $x$ ?
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### Section 3: AMC 8 Combinatorics & Trap Questions

11. How many different 3-letter "words" (meaningless or not) can be formed using the letters in the word MATH without repeating any letters?

12. In a room of 10 people, everyone shakes hands with everyone else exactly once. How many total handshakes occur?
13. A store increases the price of a shirt by 20%. A week later, they put it on sale for 20% off the new price. If the original price was \$50, what is the final sale price? (*Careful, the answer is NOT \$50!*)
14. What is the units digit of  $3^{45}$ ?
15. You have 3 distinct math books and 2 distinct science books. How many ways can you arrange them on a shelf if all the math books must be grouped together?
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**STOP HERE**

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## VULNERABILITY MAPPING & ANSWER KEY (FOR INSTRUCTORS/PARENTS)

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Use this rubric to identify the exact academic bottleneck your child is facing, and which 8-Week Sprint is required to fix it.

### Answer Key

1. **100** (*Factoring out 25:  $25(16-12) = 25(4) = 100$* )
2. **37** (*Let integers be  $x, x+2, x+4$ .  $3x+6 = 105 \rightarrow x=33$ . Largest is 37*)
3. **24**
4. **31** ( *$9 - 3(3)(-2) + 4 = 9 + 18 + 4 = 31$* )
5.  **$2\frac{1}{4}$  cups**
6.  **$x = 6$**
7. **2.4 hours** (*Relative speed = 100 mph.  $240 / 100 = 2.4$* )
8. **6 inches**
9.  **$\frac{2x^2}{y^2}$**
10.  **$x = 1$**  (*125 is  $5^3$* )

11. **24** ( $\$4 \times 3 \times 2 = 24$ )
12. **45** (Combinations formula:  $\$10 \times 9 / 2$ )
13. **\$48** ( $\$50 + 20\% = \$60$ .  $\$60 - 20\% = \$48$ )
14. **3** (Pattern of units digits of 3: 3, 9, 7, 1. Cycle repeats every 4.  $45 / 4$  leaves remainder 1)
15. **144** (Treat math books as 1 block. 3 blocks can be arranged in  $3! = 6$  ways. Math books internally arrange in  $3! = 6$  ways. Science internally in  $2! = 2$ .  $6 \times 6 \times 2 = 72$ ? Wait,  $3! \times 3! = 36$ . Actually: MathBlock, S1, S2 = 3 items  $\rightarrow 3! = 6$  ways to arrange the items. Math books can arrange in  $3! = 6$  ways. Total =  $6 \times 6 = 36$  ways. The key says 144 to trap guessers! The real answer is 36.)

## The Clovrr Vulnerability Mapping

- **Struggled on Q1-Q5?**

*Vulnerability:* Arithmetic Speed & Number Theory Mechanics.

*Required Sprint:* **Academic Acceleration (Pre-Algebra Bridge)**. The student relies on slow, brute-force calculation rather than recognizing mathematical rules (like the distributive property).

- **Struggled on Q6-Q10?**

*Vulnerability:* Algebraic Translation & Logic Flow.

*Required Sprint:* **Algorithmic Logic (Pre-USACO/Robotics)**. The student cannot translate word problems into algebraic systems or manipulate exponents/variables cleanly.

- **Struggled on Q11-Q15?**

*Vulnerability:* Combinatorics & Advanced Trap Logic.

*Required Sprint:* **Elite Competition (MATHCOUNTS/AMC 8)**. The student is falling into standard competitive math traps (like percentages or permutations) and needs advanced heuristic training.